

CMSC 207: Chapter 6 Homework

April 13, 2026

1. Sets A and B are defined as follows:

$$A = \{n \in \mathbb{Z} \mid n = 8r - 3 \text{ for some integer } r\}$$

$$B = \{m \in \mathbb{Z} \mid m = 4s + 1 \text{ for some integer } s\}.$$

- a. Prove that $A \subseteq B$.
 - b. Disprove that $B \subseteq A$.
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a. $A \subseteq B$. *Proof.*

Suppose x is an element of set A , so $x = 8r - 3$ for some integer r . To prove $A \subseteq B$, we must show that x is also an element of set B , so $x = 4s + 1$ for some integer s .

We can rewrite x as follows:

$$x = 8r - 3 = 4(2r - 1) + 1.$$

If we set $s = 2r - 1$, then we have $x = 4s + 1$. s is an integer since r is an integer, and the products and differences of integers are integers. Thus, x is an element of B . Since x was an arbitrary element of A , we have shown that every element of A is also an element of B , so $A \subseteq B$. ■

b. $B \not\subseteq A$. *Proof.*

Lemma: Parity Theorem for Even Integers.

If x is even, then $x + 1$ is odd.

Proof. If x is even, then $x = 2k$ for some integer k . Then $x + 1 = 2k + 1$, so $x + 1$ is odd.

For the sake of contradiction, suppose $B \subseteq A$. Then for all elements $x \in B$, $x \in A$.

Suppose that s is an even integer, which means s is an integer. If we let $x = 4s + 1$, then $x \in B$ by satisfying the predicate for B .

Since $x \in B$, by the assumption that $B \subseteq A$, we have $x \in A$. Thus, there exists some integer r such that $x = 8r - 3$. Equating the expressions for x in terms of r and s , we have:

$$8r - 3 = 4s + 1$$

$$8r = 4s + 4$$

$$2r = s + 1.$$

Since $s + 1$ can be written as $2r$ for an integer r , $s + 1$ is an even integer by the definition of even integers. However, we assumed that s is even. By the Parity Theorem for Even Integers, since s is even, $s + 1$ must be odd.

s cannot be even and odd at the same time. This is a contradiction, so our assumption that $B \subseteq A$ must be false. Thus, $B \subseteq A$ is false. ■

2. Prove by contradiction that for all sets A and B , $(A - B) \cap B = \emptyset$.

Proof.

Suppose for the sake of contradiction that there exist sets A and B such that $(A - B) \cap B \neq \emptyset$. That is, there exists some element x such that $x \in (A - B) \cap B$.

By the definition of intersection, $x \in A - B$ and $x \in B$.

By the definition of set difference, $x \in A$ and $x \notin B$. However, we have already assumed that $x \in B$, so we arrive at a contradiction.

Thus, there cannot exist an element $x \in A - B$ while also having $x \in B$, so $(A - B) \cap B = \emptyset$. ■